14 Year-end review and consolidation of Seconde learning

Sheet 80 Mon 14 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick $A(-1,4)$ and $B(3,1)$. Coordinates of $\overrightarrow{AB}$?
2. Core · Method Equation of (AB) in gradient-intercept form with $A(0,1)$ and $B(2,5)$.
3. Core + · Method $A(1,0)$, $B(4,1)$, $C(5,4)$. Find D so that $ABCD$ is a parallelogram.
4. Review · Quick Calculate $-5+8\times(-2)$.
5. Thread · Statement → formula The area of a square of side x is a function A of x. Give $A(x)$, a table of values for $x=1\,;\,2\,;\,3$, and say whether A is proportional to x.

Sheet 81 Tue 15 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick Write 0.00045 in standard form.
2. Core · Quick Find $[-2,3)\cap(0,5]$.
3. Core + · Method Solve $|x-1|<4$ and give S as an interval.
4. Review · Expand Expand and simplify $(2x-1)^2$.
5. Thread · Quick A cylinder has radius 3 cm and height 5 cm. Calculate its exact volume.

Sheet 82 Thu 17 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick $f(x)=x^2-4x$. Calculate $f(3)$.
2. Core · Quick Solve $x^2=9$ and then $|x|=2$.
3. Core + · Method $f(x)=2x-3$ and $g(x)=x+1$. Solve $f(x)<g(x)$.
4. Review · Quick A tap delivers 18 L in 4 min. What volume does it deliver in 10 min?
5. Thread · Reading code What is the final value of `n`?

```
n = 0
while n*n < 30:
    n = n + 1
```

Sheet 83 Mon 21 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick Mean of 5, 7, 9, 11?
2. Core · Method Data set $1\,;\,3\,;\,4\,;\,6\,;\,8\,;\,9\,;\,11$: find Q_1, the median and Q_3.
3. Core + · True/false True or false? Justify. Adding 10 to every value of a data set leaves its standard deviation unchanged.
4. Review · Reflection How many preimages do -1 and 4 have for $f(x)=x^2$?
5. Thread · Prove Show that for every real x: $(x-1)(x+1)+1=x^2$.

Sheet 84 Tue 22 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick Expand and simplify $(2x-3)(x+4)$.
2. Core · Quick Solve $3x-7\leqslant 2x+5$.
3. Core + · Method Solve $(x-1)(x+4)=0$ and then $x^2=5$.
4. Review · Reflection Membership: 5 € + 3 € per entry. Is the price proportional to the number of entries?
5. Thread · Quick A cone has radius 3 cm and height 4 cm. Calculate its slant height and its exact volume.

Sheet 85 Thu 24 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick Which multiplier corresponds to an increase of 8%?
2. Core · Method An increase of 25% is followed by a decrease of 20%. What is the overall rate of change?
3. Core + · Quick A price goes from 5,000 to 5,750 Ar. Rate of change?
4. Review · Recognise What is the cross-section of a cylinder by a plane containing its axis? By a plane perpendicular to its axis?
5. Thread · Problem Three times a number, minus 4, is equal to twice that number plus 5. Write an equation and then find the number.

Sheet 86 Mon 28 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick Direction of variation of $x \mapsto -3x + 2$?
2. Core · Method Minimum and maximum of $f(x) = x^2$ on $[-3, 1]$?
3. Core + · Method For $x > 0$, compare $\dfrac{1}{x}$ and $\dfrac{1}{x+1}$.
4. Review · Construction Describe in three steps the construction of the reflection A' of A in a line (d).
5. Thread · Method A point I satisfies $IA = IB$. Where does it lie? Which property is used?

Sheet 87 Tue 29 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick $P(A) = 0.28$. Calculate $P(\overline{A})$.
2. Core · Method 100 people, including 60 women; 6 women and 4 men are left-handed. Calculate $P_F(G)$ and $P_G(F)$.
3. Core + · Method $P(A) = 0.4$, $P_A(B) = 0.5$, $P_{\overline{A}}(B) = 0.2$. Calculate $P(B)$.
4. Review · Problem Volume of a cube of edge x: $V(x) = x^3$. Calculate $V(3)$ and find x such that $V(x) = 64$.
5. Thread · Quick In which quadrant (numbered I to IV anticlockwise) does $M(a, b)$ lie, with $a < 0$ and $b > 0$?

Sheet 88 Thu 1st July 2027 *Year-end review and consolidation*

1. Warm-up · Quick Hypotenuse of a right-angled triangle with sides 5 and 12?
2. Core · Quick Midpoint of $[AB]$ with $A(-3, 2)$ and $B(5, -4)$?
3. Core + · Write-up Show that $A(0, 0)$, $B(2, 1)$, $C(-1, 2)$ form an isosceles right-angled triangle at A.
4. Review · Quick Give an approximate value of $\dfrac{1}{3}$ correct to 10^{-2}. Is this number a decimal?
5. Thread · Problem What is the smallest integer that leaves remainder 1 when divided by 4 and by 6, other than 1?